

Albion Shoshi

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EDUCATION

The Pennsylvania State University

Aug 2022 – May 2026

B.S. in Data Science, Minor in Mathematics · GPA: 3.2 / 4.0

Relevant Coursework: Data Structures & Algorithms, Machine Learning & Algorithmic AI, Artificial Intelligence, Linear Programming, Probability Theory, Game Theory, Dynamic Programming, Object-Oriented Programming

TECHNICAL SKILLS

Programming: Python, SQL, C++, Java, C#, JavaScript

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, FinBERT, Facebook Prophet, NLP, Deep Neural Networks, Reinforcement Learning

Data & Big Data: Pandas, NumPy, Apache Spark, Hadoop MapReduce, MongoDB, Data Pipelines (ETL), Matplotlib

Tools & Frameworks: Git, Jupyter, OpenAI Gymnasium, React.js, Next.js, Unity

PROJECTS

Earnings Call Risk & Confidence Analyzer (ERCA) - Senior Capstone

Jan 2026 - May 2026

Python, FinBERT, NLP, Time-Series ML

- Built an ML pipeline that flags mispriced 0DTE options by quantifying divergence between live retail sentiment and market-implied volatility.
- Scored earnings-call and social transcripts with FinBERT and modeled social activity as a Hawkes self-exciting process to detect imminent implied-volatility crushes.
- Dynamically selected the best-performing model (Neural CDE, Multi-Transformer, Bi-Transformer, SVR) by market regime; delivered as a 5-person team.

Warden - Personal Finance Forecasting App

Aug 2025 - Present

Python, TypeScript, Next.js, TailwindCSS, SQL, Facebook Prophet

- Designed and built an envelope-style budgeting app with long-term financial forecasting and investment tracking.
- Built an ETL-style data pipeline with Facebook Prophet that ingests JSON from the database, transforms and evaluates it, and serves predictions back to the backend solving cold-start forecasting with limited user data.

NBA Talent Prediction - Research Paper

Aug 2025 - Dec 2025

Python, scikit-learn, XGBoost

- Engineered features from 20 years of NBA player statistics and clustered players into 5 star-potential tiers, then trained and evaluated Logistic Regression, XGBoost, and Random Forest models to project young players' career trajectories.

NFL Standings Prediction - Research

Aug 2024 - Dec 2024

R, Multiple Linear Regression

- Modeled team win percentage from point differential, total yardage, and defensive metrics; validated against actual 2023-season standings.

Neurosis - Horror Game

Aug 2023 - Present

Unity, C#, Blender, Steam

- Co-developed a single-player psychological-horror game exploring identity; designed the storyline, player movement, and 3D models.

Reinforcement-Learning Game Agents

Python, OpenAI Gymnasium, Pygame

- Built RL agents (Q-Learning, MDPs, DNNs) for chess, blackjack, and tic-tac-toe; the chess agent beat a 700-rated Chess.com player and the blackjack agent averaged a 45% win rate.

EXPERIENCE

The Lost Draft - Barista New York, NY

Jun 2023 - Aug 2023

- Collaborated with the team to design and demo seasonal drink menus and managed daily inventory of beans and pastries in a high-volume café.

LEADERSHIP

Computer Science Club - Secretary

Sep 2023 - May 2024

- Organized professional-development and technical events for 100+ members, growing engagement across the CS department.